Larynx & trachea

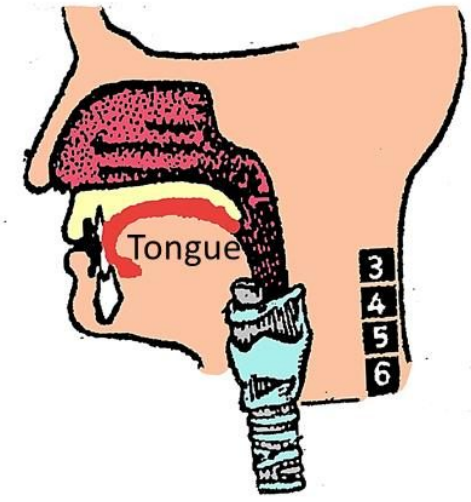
Framework, joints, membranes, compartments and spaces, the vocal fold, muscles, nerves, the paediatric larynx and the trachea.

1. Site and relations
2. Cartilages
3. Joints
4. Membranes and ligaments
5. Laryngeal inlet and cavity
6. Subsites, mucosa and lymphatics
7. Spaces of the larynx
8. The vocal fold
9. Intrinsic muscles
10. Nerve supply
11. Paediatric larynx
12. Trachea and bronchi
13. In theatre
14. On imaging

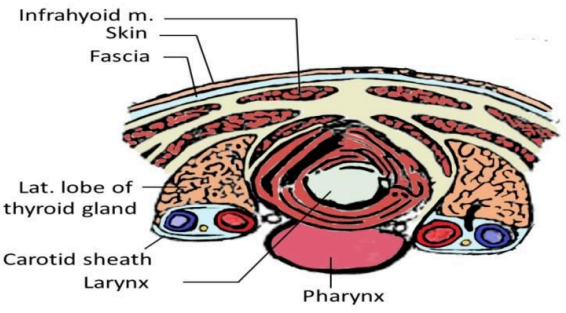
01 Site and relations

- In the front of the neck from the root of the tongue to the trachea, opposite **C3 to C6**.
- Moves up and forward in swallowing and phonation; passive side-to-side movement gives normal **laryngeal crepitus**.
- **In front:** skin, fascia, infrahyoid muscles, pyramidal lobe. **On each side:** thyroid lobe and carotid sheath. **Behind:** the laryngopharynx and piriform fossae.

CLINICAL PEARL Loss of laryngeal crepitus suggests a postcricoid or retropharyngeal mass.



Site of the larynx: C3 to C6
Dr. Sameh Doss handout, p. 170



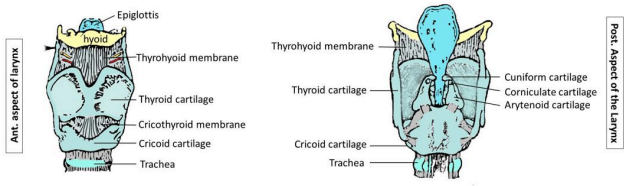
Relations of the larynx in cross-section
Dr. Sameh Doss handout, p. 170

02 Cartilages

- **Unpaired:** thyroid, cricoid, epiglottis. **Paired:** arytenoid, corniculate (Santorini), cuneiform (Wrisberg).
- **Hyaline** (ossify, starting in the thyroid at about 25 years): thyroid, cricoid, arytenoid body. **Elastic** (never ossify): epiglottis, corniculate, cuneiform, tip and vocal process of arytenoid.
- **Thyroid:** largest; two laminae meeting at **90° in men, 120° in women**. Superior notch, oblique line, superior and inferior horns. The vocal cords attach at the middle of the thyroid angle.

- **Cricoid:** the only **complete ring**; narrow arch in front, tall lamina behind. Lower border at **C6**.
- **Epiglottis:** leaf-shaped; the **petiole** attaches to the thyroid angle; the **hyoepiglottic ligament** divides it into suprahyoid and infrahyoid parts.
- **Arytenoid:** pyramid with a base on the cricoid, **vocal process** in front (vocal ligament), **muscular process** laterally (PCA, LCA), apex carrying the corniculate.

CLINICAL PEARL Ossified cartilage can look like tumour invasion on CT; the epiglottis never ossifies.



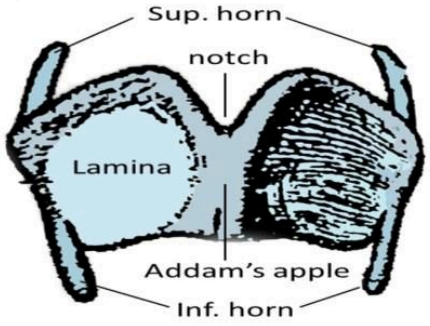
Anterior and posterior aspects of the larynx

Dr. Sameh Doss handout, p. 170



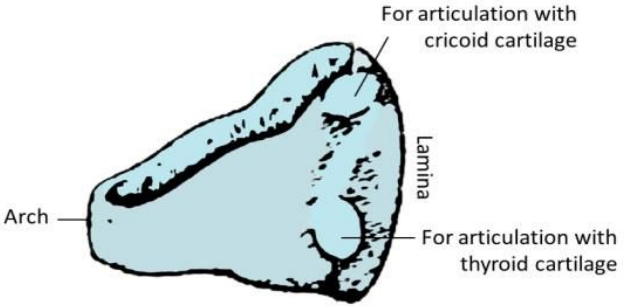
Epiglottis

Dr. Sameh Doss handout, p. 171



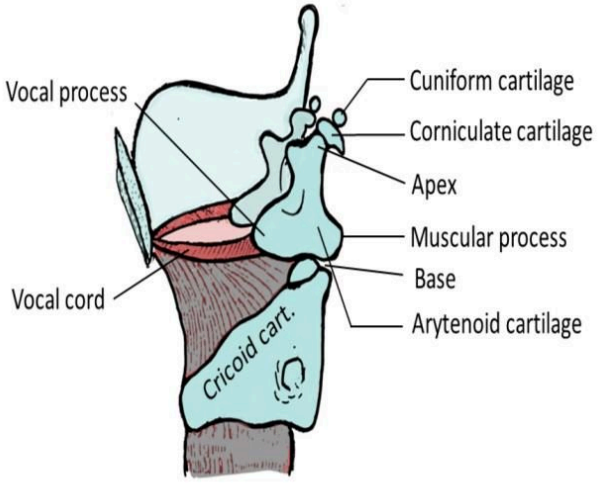
Thyroid cartilage: laminae, notch and horns

Dr. Sameh Doss handout, p. 171



Cricoid cartilage: arch and lamina

Dr. Sameh Doss handout, p. 171



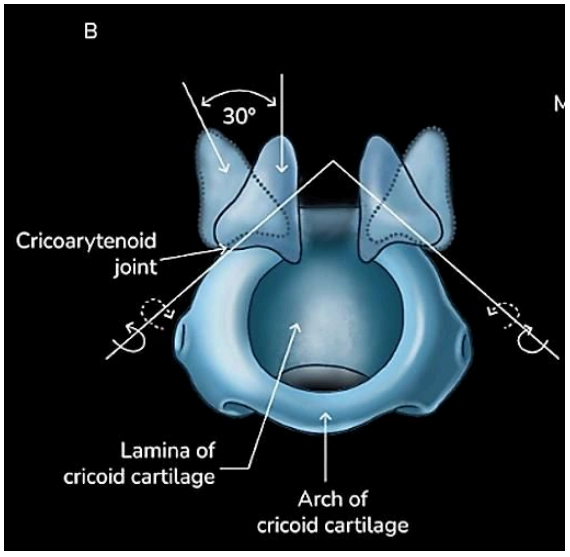
Arytenoid cartilage with corniculate and cuneiform

Dr. Sameh Doss handout, p. 171

03 Joints

- **Cricoarytenoid joint:** synovial, between the arytenoid base and a facet on the upper border of the cricoid lamina. **Rotation** about a vertical axis abducts and adducts the cords; **gliding** opens and closes the back of the glottis.
- **Cricothyroid joint:** synovial, between the inferior horn of the thyroid and the side of the cricoid. Rocking tilts the thyroid forward to lengthen the cords.

> **CLINICAL PEARL** Rheumatoid arthritis and intubation trauma can fix the cricoarytenoid joint; it looks like palsy, but the arytenoid does not move passively on palpation.



Cricoarytenoid joint

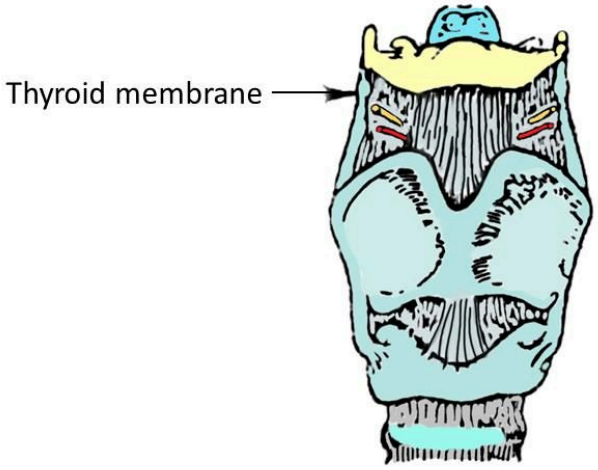
ENT Steps Anatomy, p. 234

04 Membranes and ligaments

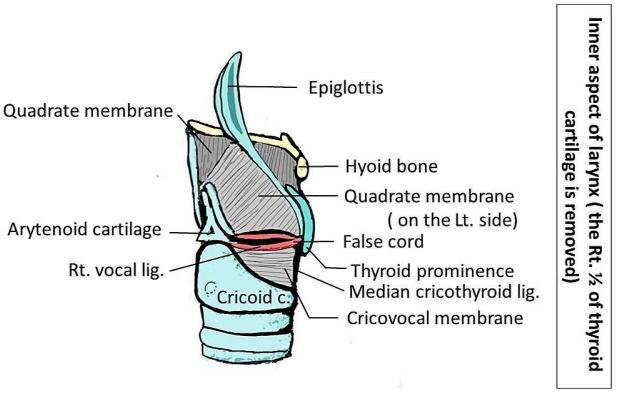
- **Extrinsic: thyrohyoid membrane** (pierced by the **internal laryngeal nerve and superior laryngeal vessels**), hyoepiglottic ligament, **cricotracheal membrane**.

- **Quadrangular membrane:** from the epiglottis to the arytenoid; upper edge in the aryepiglottic fold, **lower free edge = vestibular ligament** (false cord).
- **Cricovocal membrane (conus elasticus):** from the cricoid arch up to its **free upper edge = vocal ligament**, between the thyroid angle and vocal process. Its thick anterior part is the **median cricothyroid ligament**.
- The conus elasticus funnels into the subglottis, where small foreign bodies can impact.

> **CLINICAL PEARL** Cricothyroidotomy goes through the median cricothyroid ligament, below the vocal cords, bypassing any obstruction above them.



Thyrohyoid membrane
Dr. Sameh Doss handout, p. 172



Membranes of the larynx, inner aspect
Dr. Sameh Doss handout, p. 172

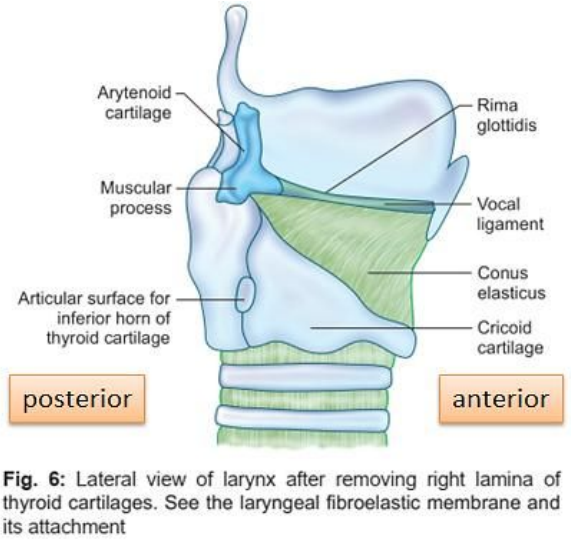


Fig. 6: Lateral view of larynx after removing right lamina of thyroid cartilages. See the laryngeal fibroelastic membrane and its attachment

Larynx after removing the right thyroid lamina: conus elasticus
ENT Steps Anatomy, p. 237

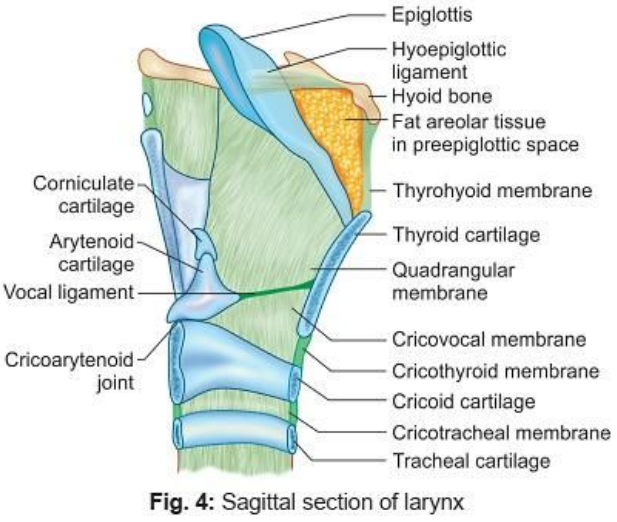


Fig. 4: Sagittal section of larynx

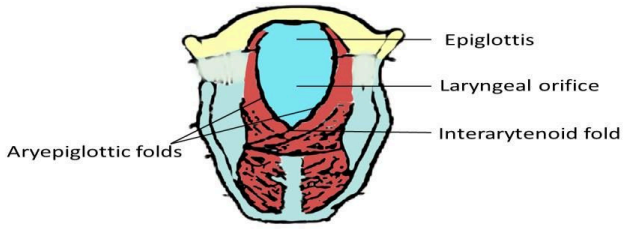
Sagittal section of the larynx: pre-epiglottic space and ligaments
ENT Steps Anatomy, p. 236

05 Laryngeal inlet and cavity

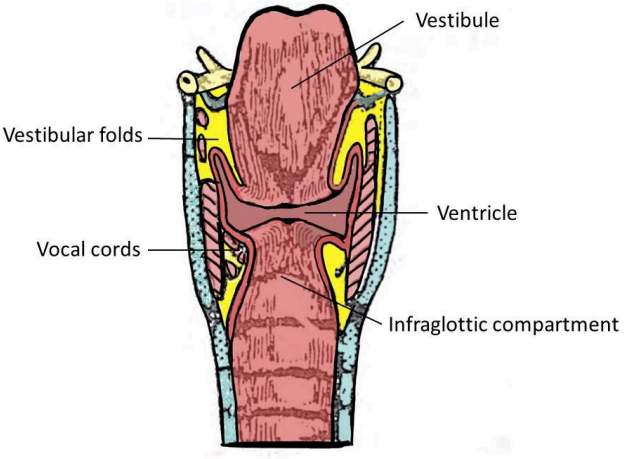
- **Inlet:** oblique; bounded by the free edge of the **epiglottis** in front, **aryepiglottic folds** at the sides and the **interarytenoid fold** behind. Closes in swallowing.
- The vestibular and vocal folds divide the cavity into the **vestibule** (inlet to false cords), **ventricle** (between false and true cords) and **infraglottic cavity** (down to the lower cricoid).

- **Saccule:** pouch from the front of the ventricle rising between the false cord and thyroid lamina; its glands lubricate the cords.
- **Rima glottidis:** narrowest part of the adult larynx. Anterior two-thirds membranous (cords), posterior third cartilaginous (vocal processes). Length about **24 mm in men, 16 mm in women**.

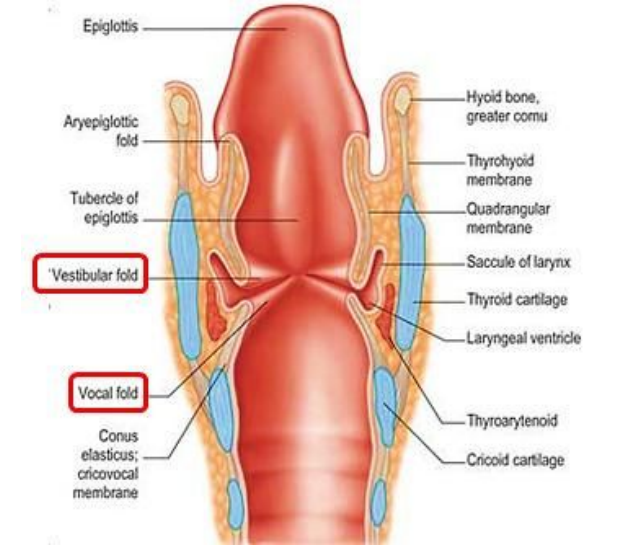
> **CLINICAL PEARL** A laryngocele is an air-filled dilated saccule; internal ones bulge the false cord, external ones push through the thyrohyoid membrane into the neck.



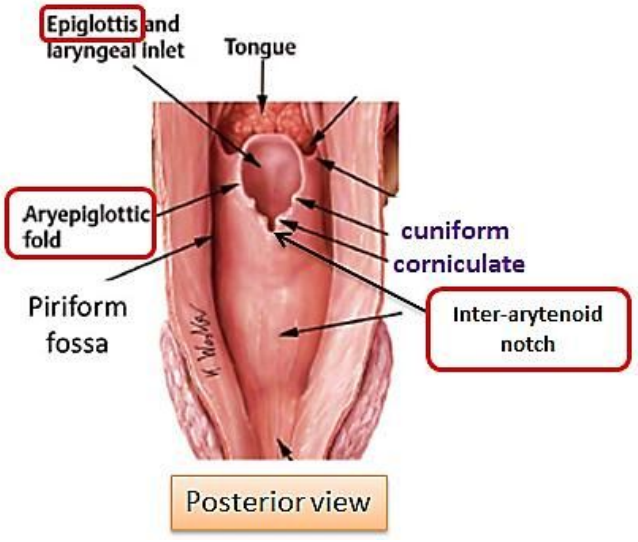
Laryngeal inlet and its boundaries
Dr. Sameh Doss handout, p. 175



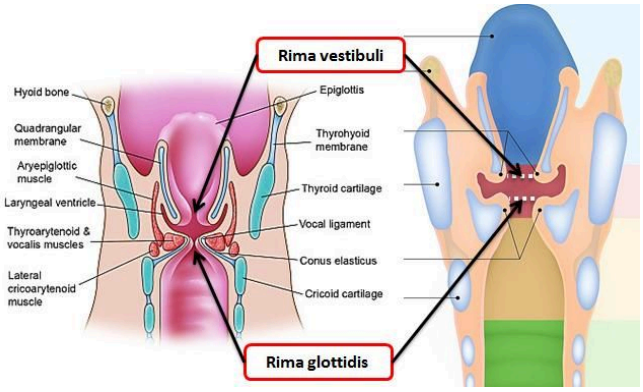
Cavity of the larynx: vestibule, ventricle, infraglottic compartment
Dr. Sameh Doss handout, p. 174



Coronal section: vestibular and vocal folds, ventricle, saccule
ENT Steps Anatomy, p. 238



Laryngeal inlet, posterior view
ENT Steps Anatomy, p. 239



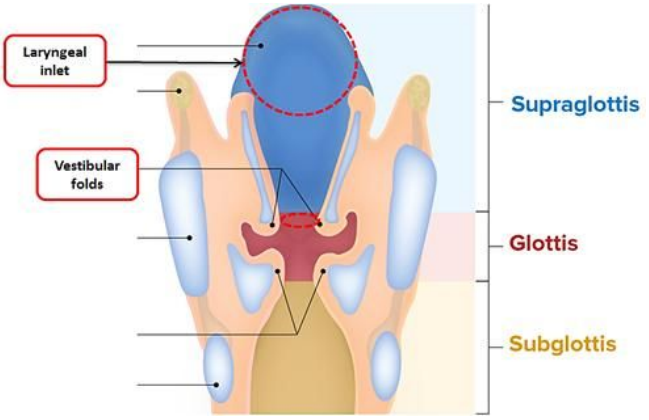
Rima vestibuli and rima glottidis

ENT Steps Anatomy, p. 241

06 Subsites, mucosa and lymphatics

- **Supraglottis:** epiglottis, aryepiglottic folds, arytenoids, false cords, ventricle. **Glottis:** true cords and anterior/posterior commissures, down to a plane 1 cm below the apex of the ventricle. **Subglottis:** down to the lower border of the cricoid.
- Epithelium is ciliated columnar except **squamous over the true cords**, upper epiglottis and aryepiglottic folds. No glands on the free edge of the cords.
- **Supraglottis:** rich, bilateral lymphatics through the thyrohyoid membrane to the **upper deep cervical (II–III)** nodes.
- **Glottis:** very sparse lymphatics.
- **Subglottis:** through the cricothyroid membrane to the **prelaryngeal (Delphian)** and pretracheal nodes, and through the cricotracheal membrane to paratracheal and lower deep cervical nodes.

> **CLINICAL PEARL** Early glottic cancer rarely spreads to nodes, so T1 glottic tumours do not need neck treatment. Supraglottic cancer often has bilateral nodes.



Supraglottis, glottis and subglottis

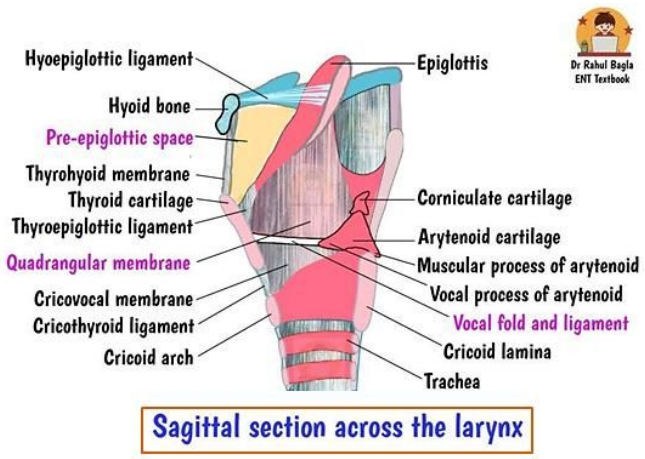
ENT Steps Anatomy, p. 239

07 Spaces of the larynx

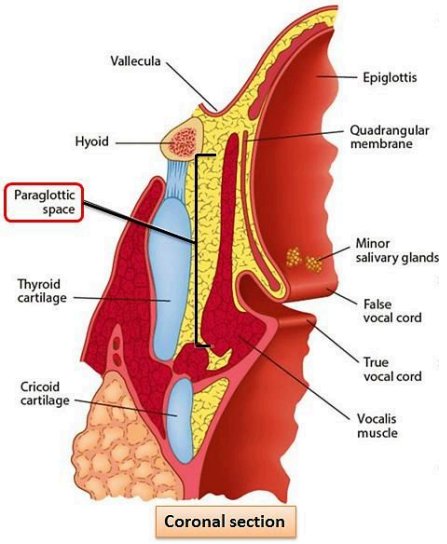
- **Pre-epiglottic space (Boyer):** fat-filled. Front: thyrohyoid membrane and upper thyroid cartilage. Above: hyoepiglottic ligament. Behind: infrahyoid epiglottis. Below: thyroepiglottic ligament. Continuous laterally with the paraglottic space.
- **Paraglottic space:** between the thyroid lamina and cricothyroid membrane laterally and the conus elasticus, ventricle and quadrangular membrane medially; behind it is the mucosa of the piriform fossa.

- **Reinke's space:** superficial lamina propria of the cord, between the epithelium and vocal ligament, limited by the arcuate lines above and below.

> **CLINICAL PEARL** Invasion of the pre-epiglottic or paraglottic space makes a laryngeal cancer T3. Paraglottic spread is how tumours become transglottic and how piriform sinus cancer reaches the larynx.



Sagittal section: pre-epiglottic space and ligaments
ENT Steps Anatomy, p. 243

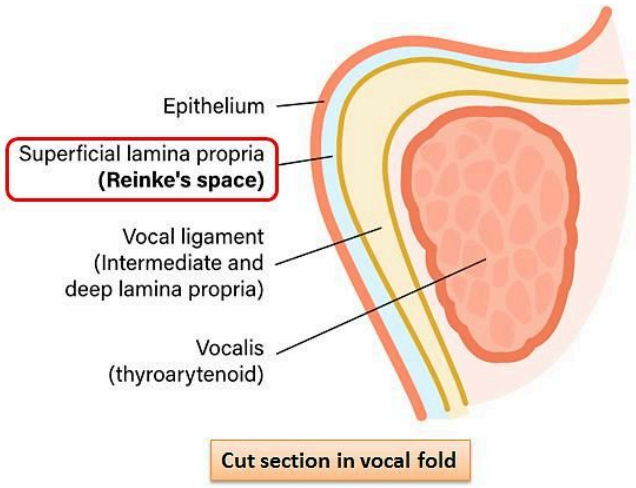


Coronal section: paraglottic space
ENT Steps Anatomy, p. 244

08 The vocal fold

- **Layers:** squamous epithelium → **superficial lamina propria (Reinke's space)** → intermediate and deep lamina propria (**vocal ligament**) → **vocalis (thyroarytenoid)**.
- **Body–cover model:** cover = epithelium + superficial lamina propria (vibrates as the mucosal wave); transition = vocal ligament; body = muscle.
- **Phase differences:** the lower edges part before the upper edges (vertical), and the front parts before the back (horizontal).
- High pitch stretches and stiffens the cover; low pitch slackens it.

> **CLINICAL PEARL** Phonosurgery stays superficial to the vocal ligament to protect the mucosal wave. Reinke's oedema is fluid in the superficial lamina propria, strongly linked to smoking.



Layers of the vocal fold: Reinke's space and vocal ligament

ENT Steps Anatomy, p. 246

09 Intrinsic muscles

- **Abductor: posterior cricoarytenoid (PCA) only**.
- **Adductors:** lateral cricoarytenoid, **interarytenoid** (the transverse arytenoid is the only unpaired intrinsic muscle; the oblique arytenoids are paired), external thyroarytenoid.
- **Tensor: cricothyroid** (tilts the thyroid forward, raising pitch). Vocalis fine-tunes tension.
- **Relaxer:** thyroarytenoid (internal part).
- **Inlet:** opened by thyroepiglottic part of thyroarytenoid; closed by the aryepiglottic and oblique arytenoid muscles.
- **Extrinsic elevators:** thyrohyoid, stylopharyngeus, salpingopharyngeus, palatopharyngeus and the suprahyoids. **Depressors:** sternohyoid, sternothyroid, omohyoid.

> **CLINICAL PEARL** "Abduct to breathe, adduct to speak." Bilateral PCA paralysis leaves both cords near the midline: good voice but stridor.

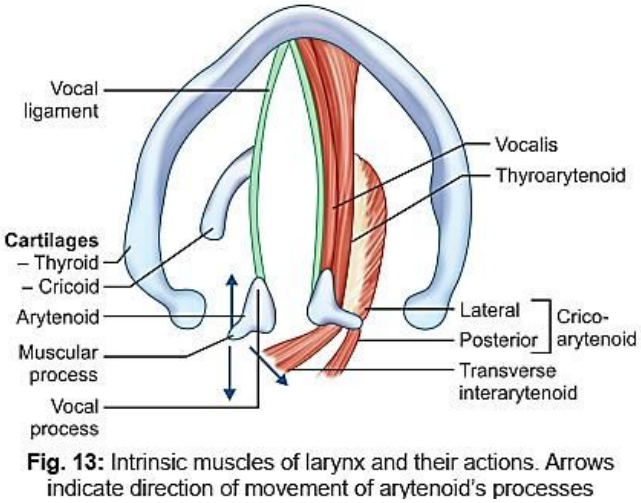
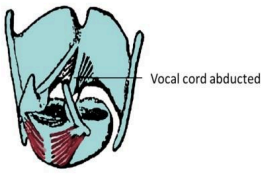
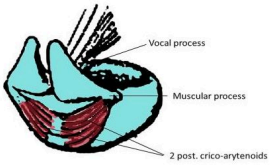


Fig. 13: Intrinsic muscles of larynx and their actions. Arrows indicate direction of movement of arytenoid's processes

Intrinsic muscles and their actions on the arytenoids

ENT Steps Anatomy, p. 249





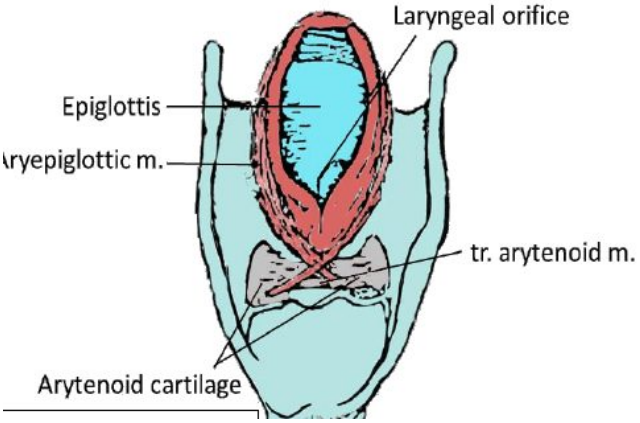
Posterior cricoarytenoids abduct the cords

Dr. Sameh Doss handout, p. 174



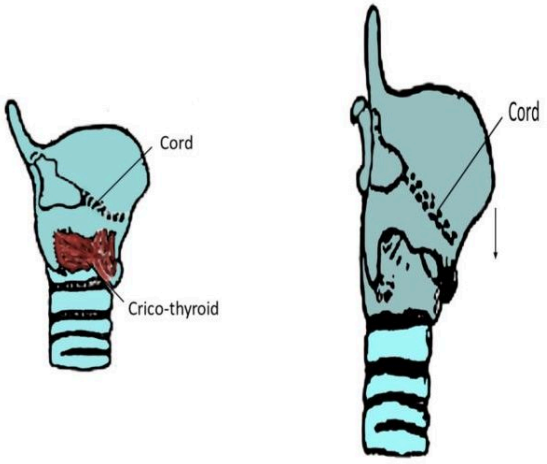
Transverse arytenoid and lateral cricoarytenoids adduct the cords

Dr. Sameh Doss handout, p. 174



Muscles that close the laryngeal inlet

Dr. Sameh Doss handout, p. 173



Cricothyroid stretches the vocal cords

Dr. Sameh Doss handout, p. 173



Thyroarytenoid relaxes the vocal cords

Dr. Sameh Doss handout, p. 173

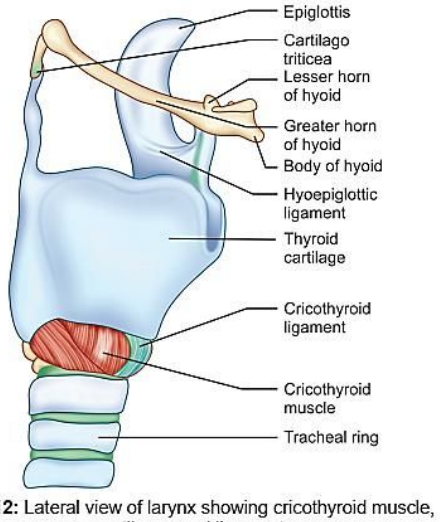


Fig. 12: Lateral view of larynx showing cricothyroid muscle, cartilages and ligaments

Cricothyroid muscle, lateral view

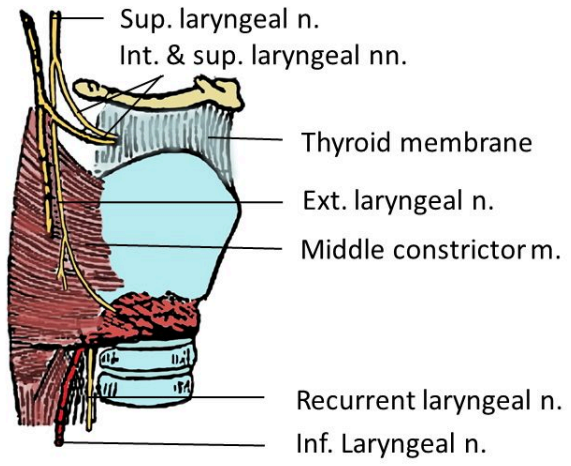
ENT Steps Anatomy, p. 249

10 Nerve supply

- All from the **vagus** (motor neurones in the nucleus ambiguus).

- **Superior laryngeal nerve** (from the inferior ganglion) runs behind the ICA and divides at the level of the hyoid's greater horn: **internal branch** pierces the thyrohyoid membrane (sensation **above the cords**); **external branch** runs with the superior thyroid artery to the **cricothyroid**.
- **Recurrent laryngeal nerve:** motor to all intrinsic muscles except cricothyroid; sensation **below the cords**. **Right** loops under the subclavian artery; **left** under the aortic arch beside the ligamentum arteriosum, so it is longer.
- RLN climbs in the tracheo-oesophageal groove and enters behind the cricothyroid joint, under the inferior constrictor.
- **Galen's anastomosis** joins the internal laryngeal and recurrent laryngeal nerves.
- **Non-recurrent laryngeal nerve:** about 0.5–1%, right side, with an aberrant right subclavian artery (arteria lusoria).

> **CLINICAL PEARL** Left cord palsy is about twice as common as right because of the long intrathoracic course (lung cancer, aortic aneurysm, cardiac surgery). External laryngeal nerve injury after thyroidectomy loses the high notes.



Nerve supply of the larynx
Dr. Sameh Doss handout, p. 175

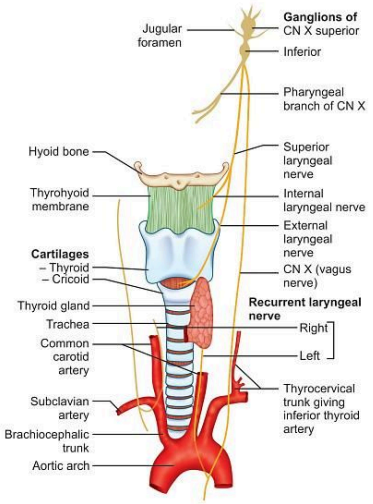
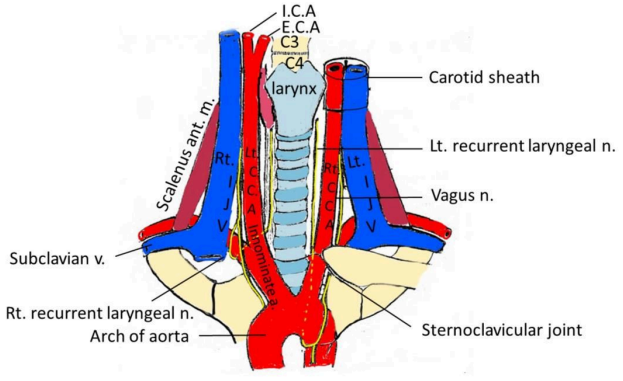


Fig. 14: Nerves supplying the larynx and their relations, especially with arteries

Nerves of the larynx and their relations
ENT Steps Anatomy, p. 251



Root of the neck: carotids, subclavian and recurrent laryngeal nerves
Dr. Sameh Doss handout, p. 135

11 Paediatric larynx

- Higher in the neck: **C3–C4** at birth versus C6 in adults.

- **Narrowest point is the subglottis (cricoid ring)**, not the glottis.
- Epiglottis omega-shaped and floppy; arytenoids relatively large; tissues looser.
- Normal term subglottis about **4.5–5.5 mm**; under 4 mm in a term baby suggests subglottic stenosis.

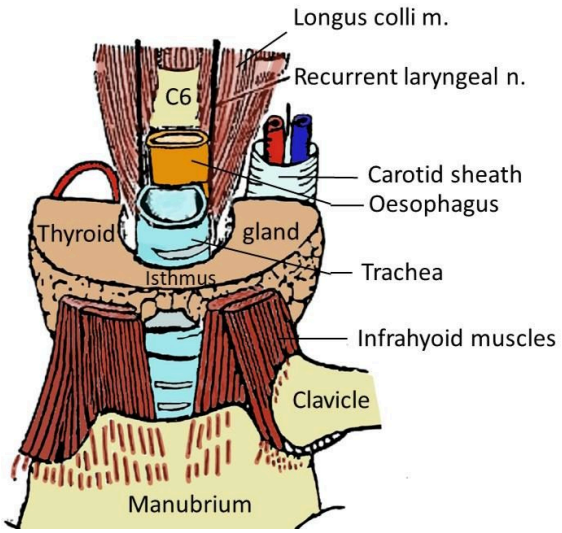
> **CLINICAL PEARL** 1 mm of oedema in a 4 mm subglottis cuts its area by about 75%: why croup and post-intubation stenosis are dangerous in infants.

12 Trachea and bronchi

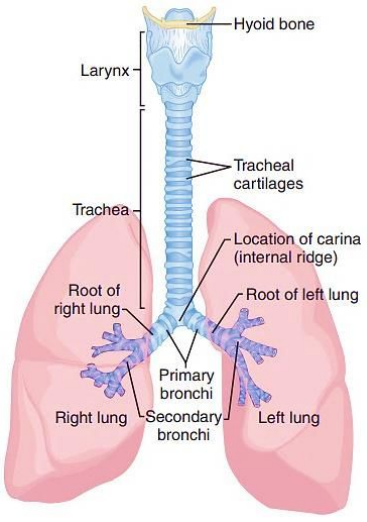
- **10–12 cm**, from the lower border of the cricoid (C6) to the **carina at T5**.
- **16–20 C-shaped hyaline rings**, open behind where the membranous wall with trachealis faces the oesophagus.
- **Cervical relations:** in front, skin, fascia, infrahyoid muscles, **thyroid isthmus over rings 2–4**, inferior thyroid veins; behind, oesophagus with the RLN in the grooves; at the sides, thyroid lobes and carotid sheath.
- **Right main bronchus** is wider, shorter and more vertical, so foreign bodies usually go right. The right upper lobe bronchus is **eparterial** (above the pulmonary artery).
- Right lung has 10 segments; left usually 8–10 (9 in Bansal/ENT Steps).

> **CLINICAL PEARL** Tracheostomy is usually placed through rings 2–4; going through ring 1 risks subglottic stenosis.

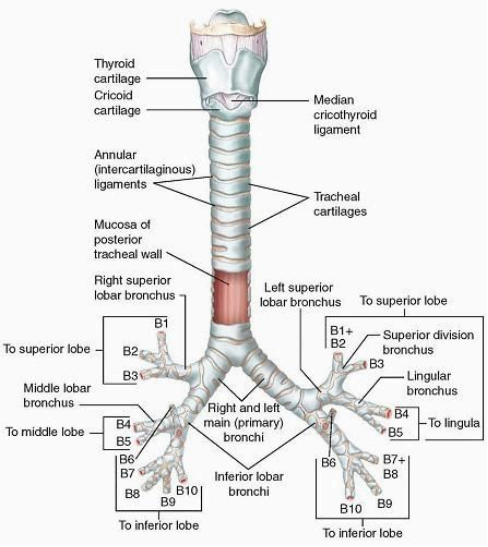
> **SOURCES DIFFER CARINA:** ENT Steps T5; cadaveric texts T4–T5. **LEFT LUNG SEGMENTS:** ENT Steps 9; Gray's 8–10.



Cervical trachea and recurrent laryngeal nerve
Dr. Sameh Doss handout, p. 176



Trachea and main bronchi
ENT Steps Anatomy, p. 253



Tracheobronchial tree, posterior view

ENT Steps Anatomy, p. 257

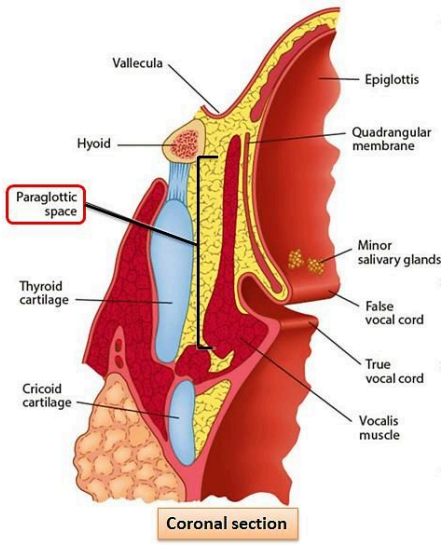
13 Surgical anatomy: In theatre

- **Cricothyroidotomy:** palpate the cricothyroid membrane, transverse stab incision in its lower part (cricothyroid arteries run along the upper border), dilate, insert a 6.0 tube.
- **Tracheostomy:** horizontal incision midway between cricoid and sternal notch, divide in the midline between the strap muscles, clamp or divide the isthmus, enter rings 2–4. Avoid ring 1 (subglottic stenosis) and going low (innominate artery).
- **Microlaryngoscopy:** stay superficial to the vocal ligament; preserve the anterior commissure mucosa on at least one side to avoid web formation.
- **Laryngectomy:** removes the hyoid to the upper tracheal rings; the internal laryngeal and superior laryngeal vessels are ligated at the thyrohyoid membrane; recurrent nerves are sacrificed.

> **CLINICAL PEARL** Children's anatomy: the cricothyroid membrane is tiny and the larynx high, so needle cricothyroidotomy is used under about 10–12 years.

14 Radiological anatomy: On imaging

- **CT with contrast** stages laryngeal cancer: look at the pre-epiglottic fat, paraglottic fat, cartilage erosion and extralaryngeal spread.
- **MRI** is more sensitive for cartilage invasion but more prone to false positives from inflammation.
- **Sclerosis of the arytenoid or cricoid** next to a tumour suggests invasion; ossification is irregular normally, so read with care.
- Normal **paraglottic fat** at the level of the false cords disappears at the true cord level, where muscle takes its place.



Coronal section: paraglottic space

ENT Steps Anatomy, p. 244

Tables

Intrinsic laryngeal muscles

Muscle	Action	Nerve
Posterior cricoarytenoid	Abducts (only abductor)	RLN
Lateral cricoarytenoid	Adducts	RLN
Interarytenoid (transverse, oblique)	Adducts; closes posterior glottis and inlet	RLN (bilateral)
Thyroarytenoid / vocalis	Adducts, shortens and relaxes; vocalis fine-tunes	RLN
Cricothyroid	Lengthens and tenses the cords (pitch)	External laryngeal nerve
Aryepiglottic	Closes the inlet	RLN

Laryngeal subsites and lymphatics

Subsite	Includes	Lymphatics	Nodes
Supraglottis	Epiglottis, AE folds, arytenoids, false cords, ventricle	Rich, bilateral	II–III via thyrohyoid membrane
Glottis	True cords, commissures, to 1 cm below the ventricle apex	Very sparse	Rarely involved early
Subglottis	To lower border of cricoid	Moderate	Delphian, pre- and paratracheal (VI), IV

Pharyngeal arch derivatives

Arch	Nerve	Skeletal	Muscles
1st	V3	Malleus (head), incus (body), Meckel's cartilage, sphenomandibular ligament	Mastication, tensor tympani, tensor veli palatini, mylohyoid, anterior digastric
2nd	VII	Stapes (superstructure), styloid, stylohyoid ligament, lesser horn and upper hyoid body	Facial expression, stapedius, stylohyoid, posterior digastric
3rd	IX	Greater horn and lower hyoid body	Stylopharyngeus

Arch	Nerve	Skeletal	Muscles
4th	X (superior laryngeal)	Thyroid cartilage	Cricothyroid, pharyngeal constrictors, levator veli palatini
6th	X (recurrent laryngeal)	Cricoid, arytenoid, corniculate	Intrinsic laryngeal muscles (except cricothyroid)

Pouches: 1st → middle ear and ET; 2nd → tonsillar fossa; 3rd → inferior parathyroid and thymus; 4th → superior parathyroid and C cells.

Vertebral levels in the neck

Level	Structure
C1	Atlas; Rouvière's node; nasopharynx level
C2	Soft palate, oropharynx begins
C3	Hyoid bone; epiglottis tip
C4	Upper border of thyroid cartilage; carotid bifurcation
C5	Vocal cords
C6	Cricoid; start of trachea and oesophagus; vertebral artery enters foramen; Chassaignac's tubercle
C7	Thoracic duct arch; vertebra prominens
T2	Lower end of retropharyngeal space
T4	Sternal angle; aortic arch
T5	Carina (T4–T5 in cadavers)
T6	Left main bronchus crosses the oesophagus (~25–27 cm from incisors)

Layers of the vocal fold

Layer	Content	Functional division
Epithelium	Stratified squamous	Cover
Superficial lamina propria (Reinke's space)	Loose, few fibres	Cover
Intermediate lamina propria	Elastic fibres	Transition (vocal ligament)
Deep lamina propria	Collagen fibres	Transition (vocal ligament)
Thyroarytenoid (vocalis)	Muscle	Body

M Mnemonics

Vocal fold movement

Abduct to breathe, adduct to speak

PCA opens the glottis; LCA, IA and TA close it

Cricothyroid

The odd one out

Only intrinsic muscle supplied by the external laryngeal nerve, and only one outside the framework

Unpaired cartilages

TEC

Thyroid, Epiglottis, Cricoid

Paired cartilages

ACC

Arytenoid, Corniculate, Cuneiform

Spaces

Pre, Para, Reinke

Pre-epiglottic and paraglottic spaces (staging T3); Reinke's space in the cord

Q Rapid recall

1. **Which intrinsic laryngeal muscle is the only abductor?** Posterior cricoarytenoid (PCA), supplied by the recurrent laryngeal nerve.
2. **Which intrinsic laryngeal muscle is NOT supplied by the RLN?** Cricothyroid, supplied by the external branch of the superior laryngeal nerve.
3. **Which intrinsic laryngeal muscle is unpaired and bilaterally innervated?** Transverse arytenoid (part of the interarytenoid; the oblique arytenoids are paired).
4. **What pierces the thyrohyoid membrane?** Internal branch of the superior laryngeal nerve and the superior laryngeal artery (and vein).
5. **Narrowest part of the infant airway?** The subglottis at the cricoid ring (in adults it is the glottis).
6. **Vertebral level of the larynx in infants vs adults?** Infant C3–C4; adult lower border of cricoid at C6.
7. **Which cartilage forms the only complete ring in the airway?** The cricoid cartilage.
8. **What is Reinke's space?** The superficial layer of the lamina propria of the vocal fold.
9. **Three lymphatic groups draining the subglottis?** Prelaryngeal (Delphian), pretracheal and paratracheal nodes (level VI).
10. **Non-recurrent laryngeal nerve: side and associated anomaly?** Right side; aberrant right subclavian artery (arteria lusoria). Incidence ~0.5–1%.
11. **Boundaries of the pre-epiglottic space?** Roof: hyoepiglottic ligament. Front: thyrohyoid membrane and thyroid cartilage. Back: epiglottis.
12. **Which laryngeal cartilages are elastic rather than hyaline?** Epiglottis, corniculate, cuneiform, and the apex and vocal process of the arytenoid.
13. **Length of the rima glottidis in men and women?** About 24 mm in men and 16 mm in women.
14. **What is the conus elasticus?** The paired cricovocal membranes; free upper edge is the vocal ligament, anterior thick part the median cricothyroid ligament.
15. **Where is the carina and how long is the trachea?** Carina at T5; trachea 10–12 cm with 16–20 C-shaped rings.
16. **Thyroid angle in men and women?** About 90° in men, 120° in women.
17. **Age at which thyroid cartilage begins to ossify?** About 25 years.
18. **Type of cricoarytenoid joint and its movements?** Synovial; rotation (abduction/adduction) and gliding.
19. **Lower free border of the quadrangular membrane?** The vestibular ligament (false cord).
20. **What is the saccule of the larynx?** A pouch from the front of the ventricle rising between the false cord and thyroid lamina; lubricates the cords.
21. **Which parts of the larynx have squamous epithelium?** True cords, upper epiglottis and aryepiglottic folds.
22. **Boundaries of the paraglottic space?** Lateral: thyroid cartilage and cricothyroid membrane. Medial: conus elasticus, ventricle, quadrangular membrane. Back: piriform mucosa.
23. **What is the body–cover model of the vocal fold?** Cover = epithelium + superficial lamina propria; transition = vocal ligament; body = vocalis.
24. **What is Galen's anastomosis?** Connection between the internal laryngeal and recurrent laryngeal nerves.
25. **Why is left RLN palsy more common?** Its longer intrathoracic course round the aortic arch.
26. **Normal subglottic diameter in a term neonate?** About 4.5–5.5 mm; under 4 mm suggests stenosis.
27. **Right lung segments vs left?** Right 10; left usually 8–10 (9 in ENT Steps).

Figures from ENT Steps Anatomy (based on Bansal) and Dr. Sameh Doss's Anatomy of Head & Neck handout, for personal study.